KS4 Maths Pathway

Target Grade:

1- I could teach this to someone else

3 - I recognise this but need help

Current Grade:

2 - I understand this

4 - I don't recognise this

		Pathway 3	Confidence 1-4	My Notes
1: Number Properties, Indices and Standard Form	1.1	Understand the terms odd, even, prime, factor, multiple, square and root. Write a number as a product of its prime factors.	1	
	1.2	Find the Highest Common Factor or Lowest Common Multiple of 2 numbers using a listing strategy and interpret these numbers in various contexts.	1	
	1.3	Simplify indices using multiplication, division and power rules.	1	
	1.4	Convert large and small numbers between ordinary and standard form. Compare and order numbers that are written in standard form. Use a calculator to perform calculations with numbers in standard form.		
2: Perimeter & Area 2D shapes	2.1	Calculate the perimeter of various 2D shapes. Use properties of 2D shapes such as isosceles triangles and rectangles etc. to find perimeter.		
	2.2	Calculate the area of various 2D shapes including rectangles, triangles, parallelograms etc. Use properties of 2D shapes to find the area.		
3: Linear Algebra	3.1	Simplify simple expressions by collecting together like terms.		
	3.2	Expand a single bracket with either a number or an algebraic term at the front.		
	3.3	Factorise single brackets with a numerical factor		
	3.4	Solve algebraic equations with an unknown on one side.		
	3.5	Form an algebraic expression from a real-world context using one or more variables.		
4: Statistical Graphs	4.1	Define the population in a study. Know the difference between population and sample. Represent, interpret and accurately construct data using Pictograms, Bar Charts (including multiple and composite) and Pie Charts. Explain why a graph is misleading.		
	4.2	Plot points on a scattergraph. Describe correlation. Draw a line of best fit and use it to estimate a value. Identify outliers.		
	4.3	Plot time series data on a graph.		
5: Formulae	5.1	Substitute positive numbers into algebraic expressions.		
	5.2	Know the difference between expressions, equations, formulae and identities.		
	5.3	Rearrange a formula that involves 1 step.		
6: Rounding, Estimation and Bounds	6.1	Round numbers to a given place value or number of decimal places		
	6.2	Estimate a result of a calculation using rounding to 1 significant figure.		
	6.3	Write down an error interval for a number rounded or truncated.		
Pythag and Trig	7.1	Use Pythagoras Theorem to calculate a missing side in a right-angled triangle		
	7.2	Recognise pythagorean triples and list examples.		
	7.3	Label a right angled triangle when given an angle. Deduce which trigonometric ratio to use.		

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7:	7.4	Label a right angled triangle when given an angle. Deduce which trigonometric ratio to use.		
8: Straight Line Graphs	8.1	Solve problems with coordinates and shapes. Plot a straight line graph on a coordinate grid. Identify the gradient and y-intercept from a written equation. Calculate the gradient of a line drawn on a grid		
	8.2	Know that parallel lines have the same gradient. Write an equation of a line that is parallel to another line.		
9: Averages	9.1	Calculate mean, median, mode and range from a list		
	9.2	Calculate the mean from a frequency table. State the mode of data in a frequency table.		
	9.3	Calculate the cumulative frequency from grouped data		
	9.4	Find upper and lower quartiles from a list of data and calculate the interquartile range		
10: Quadratic Algebra	10.1	Expand a pair of double brackets. Factorise and solve quadratic equations where all terms are positive		
	10.2	Write a quadratic expression in completed square form (where b is an even number)		
	10.3	Use the quadratic formula given values of a,b and c.		
	10.4	Use a table of values to plot		
11: Fractions, Decimals and Percentages	11.1	Simplify fractions and convert between improper fractions and mixed numbers		
	11.2	Multiply and divide fractions. Find a fraction of an amount		
	11.3	Add and subtract fractions		
	11.4	Multiply and divide by decimals using non-calculator methods. Convert between Fractions, Decimals, and Percentages.		
	11.5	Calculate the percentage of an amount. Increase/decrease a percentage by an amount. Calculate a percentage change.		
	11.6	Calculate with simple interest		
12: Angles & Circle Theorems	12.1	Know and use the following angle facts: Angles on a straight line Angles around a point Angles in a triangle Angles in a quadrilateral Vertically Opposite angles		
	12.2	Use a formula to find the total angle sum for any polygon. Use the exterior angle sum for any polygon is 360.		
	12.3	Recognise and label different parts of a circle.		
	12.4	Recognise and label different parts of a circle.		
	12.5	Recognise and label different parts of a circle.		
13: Probability	13.1	Understand that probability is a measure of chance, going from 0 "Impossible" to 1 "Certain", marking on a scale. Calculate basic probability.		
	13.2	Systematically list all different outcomes.		
	13.3	Interpret set notation and the universal set to complete a Venn diagram for two different sets, including the intersection.		
	13.4	Recognise mutually exclusive events cannot happen at the same time, and recognise this as a Venn diagram with no intersection.		
	13.5	Create and complete frequency trees, using them to solve problems involving frequency.		
Ratio & Proportion	14.1	Find the ratio of quantities in the form a : b and simplify them. Find the ratio of quantities in the form 1 : n. Identify equivalent ratios. Interpret a ratio of two parts as a fraction of a whole.		
	14.2	Use a ratio to calculate quantities given either the total amount, one amount or the difference between 2 amounts		
	14.3	Solve simple direct proportion problems (including recipe and best buy questions) by multiplicative scaling.		

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		Pathway 3	Confidence 1-4	My Notes
14:	14.4	Solve simple inverse proportion problems using scaling method. Recognising inverse proportion graphs.		
15: 3D Shapes	15.1	Identify the different nets of 3D shapes. Find the surface area of a cuboid.		
	15.2	Find the volume of a cuboid, prism or cylinder.		
16: Loci, Scale Drawings & Bearings	16.1	Construct a perpendicular bisector. Construct an angle bisector.		
	16.2	Draw loci around a point. Draw a path equidistant from two points.		
	16.3	Measure a distance on a scale drawing.		
	16.4	Measure and draw bearings of acute and obtuse angles.		
17: Plotting Further Graphs	17.1	Recognise Reciprocal, Exponential and Polynomial graphs and their corresponding equations.		
	17.2	Recognise Circle graphs and their corresponding equations.		
	17.3	Recognise the sine, cosine and tan graphs		
18. Simultaneous Equations	18.1	Solve linear simultaneous equations through elimination.		
	18.2	Use a graphical method to solve non-linear simultaneous equations where both graphs are already drawn		
19: Measurements and Real Life Graphs	19.1	Calculate speed, distance and time where the time is given as whole hours.		
	19.2	Calculate mass, density and volume.		
	19.3	Use a straight line graph to convert between different units or currencies.		
	19.4	Interpret distance-time graphs. Understand that speed is given by the gradient of the line.		
	19.5	Calculate the gradient of a line drawn on a graph.		
	19.6	Calculate the area of a triangle or a trapezium.		
20: Inequalities	20.1	Solve linear inequalities in one variable, expressing solutions on a number line.		
	20.2	Represent a linear inequality on a coordinate grid		
21: Histograms	21.1	Calculate the frequency density from a grouped frequency table		

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		Pathway 3	Confidence 1-4	My Notes
22: Plans & Elevations/ 3D Trig	22.1	Construct plans and elevations of simple 3D solids, and representations (e.g. using isometric paper) of solids from plans and elevations.		
23: Sequences & Iteration	23.1	Generate terms of a sequence using a provided rule. Recognise triangular, square and cube numbers. Identify if a sequence is linear/arithmetic, geometric, quadratic or Fibonacci		
	23.2	Carry out an iterative process a fixed number of times. Prove that an equation has a solution between 2 values by using a change of sign method.		
24: Surds	24.1	Know the difference between irrational and rational numbers.		
	24.2	Find the conjugate of an expression.		
25: Non Right Angled Trig	25.1	Substitute values into 0.5abSinC to find the area of a triangle.		
	25.2	Substitute values into a/SinA = b/SinB to calculate a length.		
	25.3	Substitute values into $a^2 = b^2 + c^2 - 2bcCosA$ to calculate a length.		
26: Similarity, Congruence & Transformations	26.1	Carry out translations, reflections, and rotations.		
	26.2	Enlarge a shape with a positive scale factor.		
	26.3	Recognise similar 2D shapes.		
	26.4	Recognise similar 3D shapes.		
	26.5	Recognise congruent shapes are identical.		
27: Functions & Graph Transformations	27.1	Use a function machine to describe a function.		
	27.2	Use function machines to describe a sequence of functions.		
	27.3	Recognise a translation when given graphs.		
	27.4	Recognise a stretch or reflection when given graphs.		
28: Algebraic Proof	28.1	Disprove a statement by finding a counter-example		
29: Vectors	29.1	Represent a 2D column vector as an arrow on a grid. Add and subtract column vectors. Multiply a column vector by a scalar quantity. Represent the results of these calculations as arrows on a grid.		

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1: Number Properties, Indices and Standard Form	1.1	Use the product of prime factors to determine if a number is a square number.		
	1.2	Find the Highest Common Factor and Lowest Common Multiple, using the product of prime factors and a Venn diagram.		
	1.3	Calculate with negative indices.		
	1.4	Add, subtract, multiply and divide numbers in standard form without a calculator.		
2: Perimeter & Area 2D shapes	2.1	Find missing quantities using a known perimeter. Find the perimeter of compound shapes. Give the circumference of a circle in exact form.		
	2.2	Calculate the area of a trapezium. Find the missing quantity of 2D shapes given the area. Find the area of compound shapes. Give the area of a circle in exact form.		
3: Linear Algebra	3.1	Simplify expressions involving multiplication and division.		
	3.2	Simplify expressions that require expanding multiple sets of single brackets.		
	3.3	Factorise a single bracket with both numerical and algebraic factors.		
	3.4	Solve algebraic equations with unknowns on boths side.		
	3.5	Set up and solve a linear equation from context.		
4: Statistical Graphs	4.1	Use a proportional method to solve problems with pie charts. Understand how bias can be introduced and how to avoid it.		
	4.2	Know the meanings of the words interpolation and extrapolation. Justify when it is appropriate to use these.		
	4.3	Plot two sets of time series data on the same graph and compare them.		
5: Formulae	5.1	Substitute negative numbers and decimals into algebraic expressions.		
	5.2	Write simple algebraic formulae involving one unknown.		
	5.3	Rearrange a formula that involves more than 1 step.		
6: Rounding, Estimation and Bounds	6.1	Round or truncate numbers to a given amount of significant figures.		
	6.2	Estimate a result of a calculation which involves dividing by a number less than 1. Use estimation in context.		
	6.3	Calculate the upper and lower bounds using numbers rounded or truncated to a known degree of accuracy.		
7: Pythag and Trig	7.1	Use Pythagoras Theorem to solve contextual problems. Use Pythagoras Theorem in shapes with multiple triangles		
	7.2	Use pythagoras' theorem to determine whether a triangle is right-angled		
	7.3	Calculate an unknown length using trigonometric ratios. Know and use exact values of sin, cos and tan for certain angles		
	7.4	Calculate an unknown angle using trigonometric ratios and inverse trigonometric functions.		

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8: Straight Line Graphs	8.1	Identify the gradient and y-intercept of a line using rearranging. Find the equation of a straight line in the form y=mx+c. Find the equation of a line given the gradient and a point. Determine whether a point lies on, below or above a line.		
	8.2	Find the equation of a line that is parallel to another.		
9: Averages	9.1	Find missing numbers in a list given 1 or more averages.		
	9.2	Calculate the estimate of the mean from a grouped frequency table. Explain why this is only an estimate.		
	9.3	Accurately plot a cumulative frequency graph		
	9.4	Accurately construct a box plot.		
10: Quadratic Algebra	10.1	Factorise and solve quadratic equations where some terms are negative. Recognise and factorise the Difference of Two Squares		
	10.2	Write a quadratic expression in completed square form (where b is an even number)		
	10.3	Use the quadratic formula, correctly identifying a,b and c.		
	10.4	Sketch a quadratic graph and label the intercepts with the axes		
11: Fractions, Decimals and Percentages	11.1	Simplify simple algebraic fractions		
	11.2	Multiply and divide mixed numbers		
	11.3	Add and subtract mixed numbers		
	11.4	Write a list of fractions, decimals and percentages in ascending or descending order.		
	11.5	Calculate the original amount after a percentage increase/decrease.		
	11.6	Calculate with compound interest		
12: Angles & Circle Theorems	12.1	Know and use angle facts with parallel lines including: Corresponding angles Alternate angles Co-Interior angles		
	12.2	Derive the interior angle sum of an n sided polygon is given by (n-2)*180. Calculate the size of an interior angle of a regular polygon.		
	12.3	Know the following Circle Theorems: The angle at the centre is double the angle at the circumference. The angle in a semicircle is a right angle. Angles in the same segment are equal.		
	12.4	Know the following Circle Theorems: Opposite angles in a Cyclic Quadrilateral add up to 180 degrees A radius meets a tangent at a right angle Tangents from the same point are equal in length		
	12.5	Know the following Circle Theorems: A radius bisects a chord at a right angle The Alternate Segment Theorem		
13: Probability	13.1	Calculate relative frequency based on the results of a fixed number of trials. Understand that as the number of trials increases, the relative frequency gets closer to the theoretical probability. Understand that more trials of an experiment produces more reliable results.		
	13.2	Systematically list all different outcomes, writing them in a sample space diagram.		
	13.3	Calculate the probability from a Venn diagram. Construct and use a two-way table to solve probability problems.		
	13.4	Recognise mutually exclusive and independent events. Understand that "OR" means add and "AND" means multiply for these events.		
	13.5	Create a Tree Diagram to represent 2 successive **independent** events. Use the tree diagram to solve problems		
14: Ratio & Proportion	14.1	Combine two ratios together. E.g. a:b and b:c to find a:b:c		
	14.2	Split a quantity into three or more parts given the ratio of the parts.		
	14.3	Complete a table of values for variables in direct proportion.		
	14.4	Complete the table of values for variables in inverse proportion.		

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		Pathway 3	Confidence 1-4	My Notes
15: 3D Shapes	15.1	Find the surface area of prisms and cylinders.		
	15.2	Find the volume of compound shapes made up of cuboids, prisms and cylinders.		
16: Loci, Scale Drawings & Bearings	16.1	Construct a perpendicular from a point to a line. Construct a perpendicular through a point on a line.		
	16.2	Draw a path of points equidistant from two sides. Shade regions that meet a variety of criteria.		
	16.3	Calculate real world distances from a scale drawing		
	16.4	Measure and draw bearings of reflex angles.		
Plottin g Further	17.1	Plot Reciprocal, Exponential and Polynomial graphs from a table of values.		
	17.2	Plot Circle graphs given their equations. Verify if coordinates lie on, inside or outside a circle.		
	17.3	Use a table of values to plot the sine, cosine or tan graph between 0 and 360.		
18. Simultaneous Equations	18.1	Solve linear simultaneous equations through elimination, where multiplication of one (or both) equation(s) is required.		
	18.2	Plot a straight line on a curve in order to solve non-linear simultaneous equations		
19: Measurements and Real Life Graphs	19.1	Calculate speed, distance and time, where the units of time are not whole hours (e.g. 2 hours, 20 minutes).		
	19.2	Calculate mass and density where you are required to calculate the volume of a shape first.		
	19.3	Understand the relationship between the gradient of a conversion graph and the conversion rate.		
	19.4	Plot distance-time graphs.		
	19.5	Calculate the average speed from a non-linear distance time graph using a gradient.		
	19.6	Split the area below a non-linear distance time graph into triangles and trapeziums.		
20: Inequalities	20.1	Solve more complex linear inequalities. e.g. 1< 3x - 5 < 10		
	20.2	Represent multiple linear inequalities on a coordinate grid		
21: Histograms	21.1	Interpret and construct diagrams for grouped data as appropriate, i.e. histograms with equal class intervals.		
22: Plans & Elevations/ 3D Trig	22.1	Construct plans and elevations of more complex 3D solids, and representations (e.g. using isometric paper) of solids from plans and elevations.		
ences & tion	23.1	Generate a sequence by spotting a pattern or using a term-to-term rule given algebraically or in words. Generate the nth term of an arithmetic sequence.		

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		Pathway 3	Confidence 1-4	My Notes
23: Sequences, Iterative Methods	23.2	Use a decimal search method to find the solution to an equation to a required degree of accuracy.		
24: Surds	24.1	Simplify surds using multiplication property and square numbers.		
	24.2	Rationalise the denominator containing a single surd.		
25: Non Right Angled Trig	25.1	Correctly label a non right angled triangle and substitute values into 0.5abSinC to calculate the area of a triangle.		
	25.2	Correctly label a non right angled triangle and substitute values into a/SinA = b/SinB to calculate a length.		
	25.3	Correctly label a non right angled triangle and substitute values into $a^2 = b^2 + c^2 - 2bcCosA$ to calculate a length.		
26: Similarity, Congruence & Transformations	26.1	Carry out translations, reflections, and rotations, and describe the result of a transformation.		
	26.2	Enlarge a shape from a point with a positive scale factor.		
	26.3	Calculate missing lengths on similar 2D shapes, using a length scale factor.		
	26.4	Calculate missing lengths on 3D shapes using a length scale factor.		
	26.5	Know that the 4 conditions for congruent triangles are either SSS, SAS, ASA, or RHS.		
27: Functions & Graph Transformations	27.1	Use a function machine to describe a function, and identify the inverse as the reverse.		
	27.2	Use function machines to describe a sequence of functions and identify this as a composite function.		
	27.3	Vertically translate a function by adding on an amount.		
	27.4	Vertically stretch a function by multiplying by a scale factor.		
28: Algebraic Proof	28.1	Understand and use the concepts and vocabulary of expressions, equations, formulae, inequalities, terms and factors.		
29: Vectors	29.1	Use algebra to describe a vector from one point to another		

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		Pathway 3	Confidence 1-4	My Notes
1: Number Properties, Indices and Standard Form	1.1	Use the product of prime factors to determine the smallest number that a number should be multiplied by in order to create a square number	1	
	1.2	Find the Highest Common Factor and Lowest Common Multiple of 2 large numbers that have already been expressed as a product of their prime factors.	1	
	1.3	Calculate with fractional indices.	1	
	1.4	Incorporate the laws of indices and factorisation to efficiently simplify and solve complex standard form calculations.	1	
2: Perimeter & Area 2D shapes	2.1	Calculate the arc length and total perimeter of a sector of a circle. Find a missing quantity given the arc length or perimeter of the sector	2	
	2.2	Calculate the area of a sector of a circle. Find a missing quantity given the area of the sector.	1	
3: Linear Algebra	3.1	Simplify algebraic expressions containing terms with multiple variables.	2	
	3.2	Expand and simplify algebraic expressions involving subtractions of brackets.	2	
	3.3	Factorise by taking out common algebraic term by using the laws of indices.	3	
	3.4	Solve an equation containing fractions or roots.	2	
	3.5	Set up and solve linear equation from context involving fractions of amounts.	2	
4: Statistical Graphs	4.1	Solve problems with pie charts involving algebra and ratios.	1	
	4.2	Solve contextual problems from scatter graphs which may also involve other areas of maths such as percentages and probability.	1	
	4.3	Identify seasonal and annual trends from time series data.	1	
5: Formulae	5.1	Understand the variables for equations of motion (SUVAT). Choose the appropriate equation to use given some values and calculate the answer.	3	
	5.2	Write an algebraic formula involving 2 unknowns.	2	
	5.3	Rearrange a formula where the subject is negative or in the denominator of a fraction.	1	
6: Rounding, Estimation and Bounds	6.1	Round or truncate numbers to a given amount of significant figures.	1	
	6.2	Estimate a result of a calculation which involves dividing by a number less than 1. Use estimation in context.	1	
	6.3	Calculate the upper or lower bound of a calculation	2	
7: Pythag and Trig	7.1	Use Pythagoras Theorem in non-triangular shapes to calculate missing sides	1	
	7.2	Recognise a pattern for pythagorean triples and use them in a variety of problems.	3	
	7.3	Solve problems which require the use of Pythagoras Theorem and Trigonometry	2	
	7.4	Solve problems which require the use of Pythagoras Theorem and Trigonometry	2	

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8: Straight Line Graphs	8.1	Find the equation of a line given two points on the line.	2	
	8.2	Know that the gradient of a perpendicular line is the negative reciprocal. Find the equation of a line that is perpendicular to another	2	
9: Averages	9.1	Solve weighted mean problems		
	9.2	Find the median or median group from ungrouped and grouped frequency tables.		
	9.3	Find estimates for the median and quartiles from a cumulative frequency graph		
	9.4	Construct a box plot from a cumulative frequency graph		
10: Quadratic Algebra	10.1	Expand triple brackets. Factorise and solve quadratic equations with a prime value of a.		
	10.2	Write a quadratic expression in completed square form		
	10.3	Recall and use the quadratic formula.		
	10.4	Factorise a given quadratic in order to plot its graph. Hence, be able to identify the vertex.		
11: Fractions, Decimals and Percentages	11.1	Simplify algebraic fractions which require factorising linear expressions	2	
	11.2	Multiply and divide algebraic fractions	2	
	11.3	Add and subtract algebraic simple fractions	2	
	11.4	Use algebra to convert a recurring decimal into a fraction	2	
	11.5	Identify the decimal multiplier for a percentage increase or decrease	2	
	11.6	Express compound percentage change using growth and decay formulae	4	
12: Angles & Circle Theorems	12.1	Apply these angle properties in more formal proofs of basic geometrical results.		
	12.2	Determine the number of sides of a regular polygon given exterior or interior angles.		
	12.3	Apply knowledge of these circle theorems in more complicated angle problems		
	12.4	Apply knowledge of these circle theorems in more complicated angle problems		
	12.5	Apply knowledge of these circle theorems in more complicated angle problems		
13: Probability	13.1	Calculate probabilities, including from sample spaces, and compare these to experiments. Recognise the relationship between expectation value, relative frequencies, and the number of trials.		
	13.2	Calculate the number of combinations using the product rule for counting.		
	13.3	Complete a 2-circle Venn diagram for given information. Understand the meanings of the Intersection (∩), Union (∪) and Complement (') symbols		
	13.4	Use P(A or B) = P(A) + P(B) - P(A and B), and that mutually exclusive events mean P(A and B) = 0. Recognise independent events, and P(A and B) = P(A) x P(B) for them.		
	13.5	Create a Tree Diagram to represent 2 successive **conditional** events. Use the tree diagram to solve problems.		
14: Ratio & Proportion	14.1	Convert a given linear equation in two variables into a ratio between variables.		
	14.2	Solve problems which involve changing ratios		
	14.3	Form equations and solve formal direct proportion problems. Recognise that if y = kx, where k is a constant, then y is proportional to x. Recognise graphs of direct proportion.		
	14.4	Form equations and solve formal inverse proportion questions Recognise that if y = k/ x , where k is a constant, then y is inversely proportional to x. Recognise graphs of inverse proportion.		
15: 3D Shapes	15.1	Find the surface area of cones and spheres. Use pythagoras theorem within a cone to find the surface area.		
	15.2	Find the volume of cones, spheres and pyramids. Use pythagoras theorem within cones and pyramids to find their volumes.		

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16: Loci, Scale Drawings & Bearings	16.1	Construct triangles given a combination of lengths and angles.		
	16.2	Draw a path that is equidistant around a shape. Shade regions that satisfy worded problems.		
	16.3	Determine the key of a scale drawing given the real world length and/or unit conversions.		
	16.4	Measure and draw bearings by applying angle properties.		
Plotting Further	17.1	Use Reciprocal, Exponential and Polynomial graphs to solve equations.		
	17.2	Find the equation of the radius of a circular graph.		
	17.3	Sketch a sine, cosine or tan graph between 0 and 360, labelling key values on both axes.		
18. Simultaneous Equations	18.1	Design a mathematical model of a real-world scenario using a pair of simultaneous equations. (e.g. supply and demand)		
	18.2	Solve non-linear simultaneous equations using a substitution method.		
19: Measurements and Real Life Graphs	19.1	Solve speed, distance and time problems which involve multi-part journeys.		
	19.2	Solve mass, density and volume problems which involve combining 2 or more objects together.		
	19.3	Draw a conversion graph from a given linear model to convert different types of measurements e.g Celsius to Fahrenheit.		
	19.4	Plot velocity-time graphs. Understand that acceleration is given by the gradient of the line.		
	19.5	Estimate the gradient of a non-linear graph by drawing a tangent and calculating the gradient of a tangent.		
	19.6	Estimate the distance travelled from a non-linear velocity-time graph by splitting the shape into triangles and trapeziums and calculating their combined area.		
20: Inequalities	20.1	Solve quadratic inequalities.		
	20.2	Identify a region on a coordinate grid that satisfies multiple linear inequalities.		
21: Histograms	21.1	Interpret and construct diagrams for grouped data as appropriate, i.e. histograms with unequal class intervals.		
22: Plans & Elevations/ 3D Trig	22.1	Use Pythagoras' theorem in 3D shapes.		
23: Sequences & Iteration	23.1	Use subscript notation for position-to-term and term-to-term rules. Find a formula for the nth term of a geometric sequence or a simple quadratic sequence.		
	23.2	Use an iterative process to find the solution to an equation. Know that each value generated by the process is an increasingly accurate estimate of the true solution.		
24: Surds	24.1	Adding and subtracting surds. Expanding single brackets containing surds.		
	24.2	Rationalise the denominator using conjugate and difference between two squares.		

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		Pathway 3	Confidence 1-4	My Notes
Non Right Angled	25.1	Use 0.5abSinC to calculate the area of a non right angled SAS triangle.		
	25.2	Use the Sine Rule to calculate missing lengths and angles in non right angled triangles.		
	25.3	Use the Cosine Rule to calculate missing lengths and angles in non right angled triangles.		
26: Similarity, Congruence & Transformations	26.1	Carry out multiple translations, reflections, and rotations, and describe the single result of multiple transformations.		
	26.2	Enlarge a shape from a point with a fractional scale factor.		
	26.3	Prove two 2D shapes are similar and calculate missing lengths using a length scale factor.		
	26.4	Calculate length, area, and volume scale factors, and understand the relationship between them.		
	26.5	Prove 2 triangles are congruent using SSS, SAS, ASA or RHS in a formal proof.		
27: Functions & Graph Transformations	27.1	Find the inverse of a linear function by rearranging algebra to make x the subject.		
	27.2	Write down a composite function by substituting algebra into another function.		
	27.3	Perform a horizontal translation on a graph. Write the equation of a graph that has been translated.		
	27.4	Horizontally stretch a function by multiplying by a scale factor. Write the equation of a stretched graph.		
28: Algebraic Proof	28.1	Recognise the difference between an equation and an identity, and show algebraic expressions are equivalent.		
29: Vectors	29.1	Use algebra to describe a vector from one point to another that involves a ratio		

KS4 Maths Pathway

Target Grade:

1- I could teach this to someone else

3 - I recognise this but need help

Current Grade:

2 - I understand this

4 - I don't recognise this

		Pathway 3	Confidence 1-4	My Notes
1: Number Properties, Indices and Standard Form	1.1	Write a number as a product of prime factors where algebra is required.		
	1.2	Find the Highest Common Factor and Lowest Common Multiple of 3 numbers using the product of prime factors and a Venn diagram.		
	1.3	Calculate with negative and fractional indices.		
	1.4	Apply standard form calculations to solve real-life problems involving speed, distance, time, population growth, and astronomical distances.		
2: Perimeter & Area 2D shapes	2.1	Solve complex problems involving the perimeter of a sector		
	2.2	Solve more complex problems involving the area of a sector of a circle.		
3: Linear Algebra	3.1	Simplify algebraic expressions containing terms involving powers and multiple variables.		
	3.2	Expand single brackets where each term contains multiple variables by using the laws of indices.		
	3.3	Factorise algebraic expressions with a bracket as a common factor.		
	3.4	Solve linear equations with fractions containing algebraic numerator and denominator.		
	3.5	Set up and solve linear equations involving percentage increase/decrease of an amount or fractions of amounts.		
4: Statistical Graphs	4.1	Solve complex problems with pie charts involving algebra and ratios.		
	4.2	Understand that correlation does not equal causation.		
	4.3	Identify seasonal and annual trends from time series data.		
5: Formulae	5.1	Use the equations of motion to solve problems in context.		
	5.2	Use knowledge of algebraic identities to work out the values of missing constants and coefficients		
	5.3	Rearrange a formula where the subject appears more than once.		
6: Rounding, Estimation and Bounds	6.1	Round or truncate numbers to a given amount of significant figures.		
	6.2	Estimate a result of a calculation which involves dividing by a number less than 1. Use estimation in context.		
	6.3	Calculate the result of a complex calculation by considering upper and lower bounds and suitable degree of accuracy.		
7: Pythag and Trig	7.1	Use algebra to prove Pythagoras Theorem		
	7.2	Derive a formula for finding pythagorean triples.		
	7.3	Apply trigonometric ratios to a variety of contextual problems involving multiple shapes.		
	7.4	Apply trigonometric ratios to a variety of contextual problems involving multiple shapes.		

KS4 Maths Pathway

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1- I could teach this to someone else

3 - I recognise this but need help

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		Pathway 3	Confidence 1-4	My Notes
8: Straight Line Graphs	8.1	Calculate the length and midpoint of a line segment. Solve problems involving points dividing line segments in a given ratio.		
	8.2	Use properties of 2D shapes to find equations of straight lines.		
9: Averages	9.1	Understand the benefits and drawbacks of each average and know how they can be affected by outliers.		
	9.2	Solve problems with frequency tables including those requiring reverse processes.		
	9.3	Find values from a cumulative frequency graph and solve problems with these values which involve other areas of maths.		
	9.4	Compare 2 box plots using their medians and interquartile ranges		
10: Quadratic Algebra	10.1	Factorise and solve quadratic equations with any integer value of a.		
	10.2	Solve a quadratic equation using the method of completing the square		
	10.3	Understand the derivation of the quadratic formula.		
	10.4	Use completing the square to identify the vertex of a quadratic graph		
11: Fractions, Decimals and Percentages	11.1	Simplify algebraic fractions which require factorising quadratic expressions		
	11.2	Multiply and divide algebraic fractions which require factorisation		
	11.3	Add and subtract complex algebraic fractions		
	11.4	Use algebra to convert a complex recurring decimal into a fraction		
	11.5	Identify the decimal multiplier for a percentage increase or decrease where the percentage is a decimal		
	11.6	Solve problems using growth and decay formulae		
12: Angles & Circle Theorems	12.1	Apply these angle properties in more formal proofs of complex geometrical results.		
	12.2	Solve complex angle problems when quantities are given in ratios or using algebra by forming linear equations.		
	12.3	Prove these circle theorems		
	12.4	Prove these circle theorems		
	12.5	Prove these circle theorems		
13: Probability	13.1	Calculate the overall relative frequency of an event across multiple trials, by combining the results of each trial and then calculating the relative frequency based on the total number of trials.		
	13.2	Calculate the number of combinations using the product rule for counting, knowing when to add or multiply different sets of combinations.		
	13.3	Complete a 3-circle Venn diagram for given information.		
	13.4	Use P(A or B) = P(A) + P(B) - P(A and B), and that mutually exclusive events mean P(A and B) = 0. Recognise independent events, and show P(A and B) ≠ P(A)XP(B) for dependent events.		
	13.5	Use algebra within a tree diagram to solve probability problems.		
14: Ratio & Proportion	14.1	Determine a ratio between two or three quantities from a given context.		
	14.2	Use algebra to solve complex ratio problems		
	14.3	Formulate equations and solve problems involving a quantity in direct proportion to a power or root of another quantity.		
	14.4	Formulate equations and solve problems involving a quantity in inverse proportion to a power or root of another quantity.		
15: 3D Shapes	15.1	Find the surface area of compound 3D shapes.		
	15.2	Find the volume of compound 3D shapes.		

KS4 Maths Pathway

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		Pathway 3	Confidence 1-4	My Notes
16: Loci, Scale Drawings & Bearings	16.1	Construct more complex shapes, angles and inscribe regular polygons.		
	16.2	Apply a number of different constructions in order to create loci and/or regions that solve contextual problems.		
	16.3	Calculate lengths and/or scale key given a ratio of m:n.		
	16.4	Measure and draw bearings by applying angle properties and trigonometry.		
17: Plotting Further Graphs	17.1	Plot and interpret Reciprocal, Exponential and Polynomial graphs in a variety of contexts.		
	17.2	Find the equation of a tangent to a circular graph		
	17.3	Use a sketch/plot of a trigonometric graph to solve an equation between a fixed range of values.		
18. Simultaneous Equations	18.1	Solve linear simultaneous equations by substitution requiring rearrangement.		
	18.2	Solve non-linear simultaneous equations using a substitution method where some rearrangement is required		
19: Measurements and Real Life Graphs	19.1	Use SUVAT equations to solve various kinematics problems.		
	19.2	Work out the density of an alloy composed of two elements.		
	19.3	Investigate why extrapolation from a conversion graph only works when units are in direct proportion and when it is appropriate to extrapolate if the units are not indirect proportion.		
	19.4	Calculate the distance travelled using the area below a velocity-time graph.		
	19.5	Interpret the gradient of a non-linear graph in context.		
	19.6	Decide whether the estimate for distance travelled from a non-linear velocity-time graph is an overestimate or an underestimate and justify your decision.		
20: Inequalities	20.1	Express solutions in set notation. e.g. {x: x>3}		
	20.2	Determine the linear inequalities required to represent a region that is drawn on a coordinate grid		
21: Histograms	21.1	Solve reverse histogram problems by determining axis scales using known frequencies.		
22: Plans & Elevations/ 3D Trig	22.1	Use right-angle trigonometry in 3D shapes.		
23: Sequences & Iteration	23.1	Find the nth term formula for a complex quadratic sequence.		
	23.2	Rearrange an equation to give a required iterative formula.		

KS4 Maths Pathway

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		Pathway 3	Confidence 1-4	My Notes
24: Surds	24.1	Expanding double brackets containing surds.		
	24.2	Write complicated calculations involving surds as a multiple of a single surd.		
25: Non Right Angled Trig	25.1	Derive and use the formula 0.5abSinC to calculate the area of a non right angled SAS triangle.		
	25.2	Derive and use the Sine Rule to calculate missing sides and angles in non right angled triangles.		
	25.3	Know when to use the Cosine and Sine Rule to calculate missing sides and angles in non right angled triangles.		
26: Similarity, Congruence & Transformations	26.1	Identify invariant points after a series of transformations have been carried out.		
	26.2	Enlarge a shape from a point with a negative scale factor.		
	26.3	Calculate missing lengths in a variety of shapes, including triangles within other shapes.		
	26.4	Use scale factors to solve similar 3D shape problems, e.g. calculating the surface area when given volumes.		
	26.5	Use congruency to prove angle and shape results.		
27: Functions & Graph Transformations	27.1	Find the inverse of a function involving quadratics, cubics and roots. Extend your knowledge of trigonometric inverse functions.		
	27.2	Write down a composite functions involving trigonometric functions and algebraic functions.		
	27.3	Perform multiple translations on the same graph.		
	27.4	Perform multiple transformations on the same graph		
28: Algebraic Proof	28.1	Use algebra to construct proofs and arguments.		
29: Vectors	29.1	Use algebra to complete a geometric proof involving parallel vectors and collinear points.		